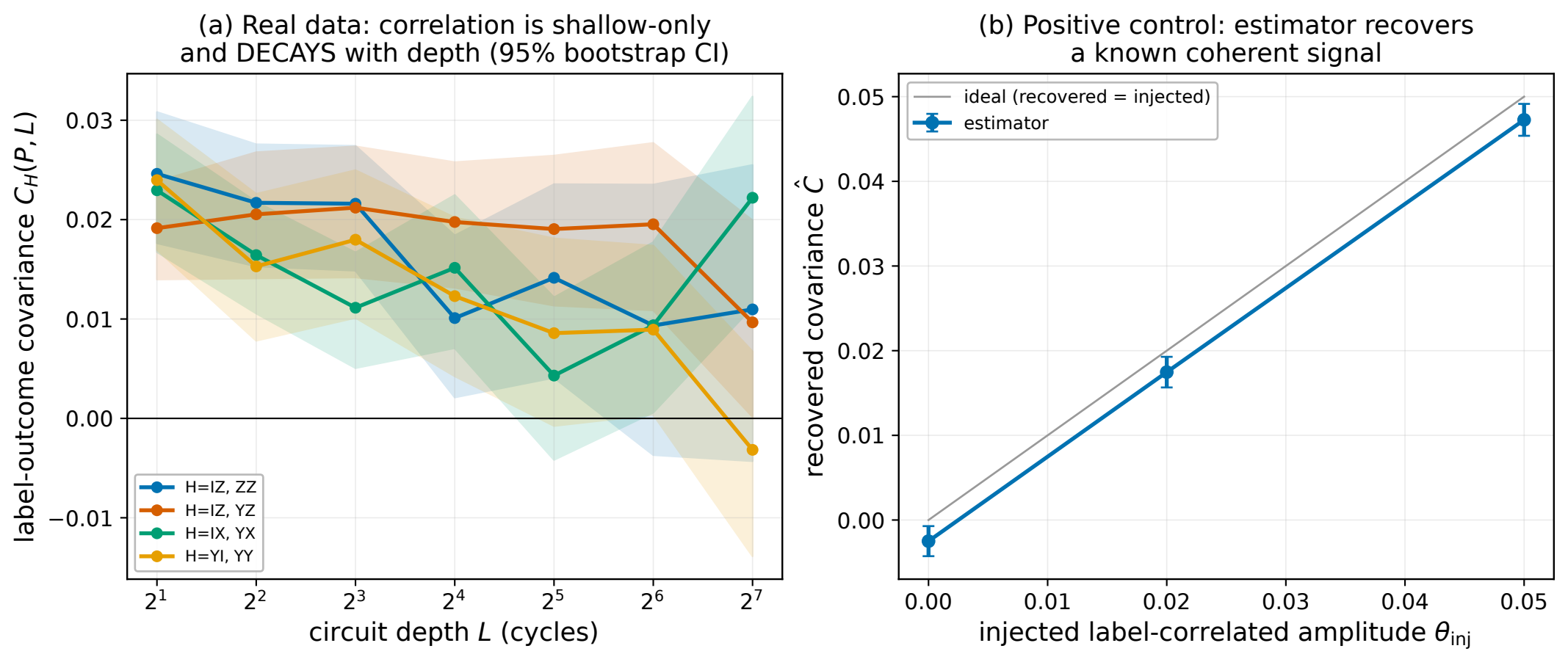
(a) Real data: correlation is shallow-only
and DECAYS with depth (95% bootstrap CI)
label-outcome covariance $C_H(P, L)$
0.03
0.02
0.01
0.00
−0.01
H=IZ, ZZ
H=IZ, YZ
H=IX, YX
H=YI, YY
2^1
2^2
2^3
2^4
2^5
2^6
2^7
circuit depth L (cycles)
(b) Positive control: estimator recovers
a known coherent signal
ideal (recovered = injected)
estimator
recovered covariance $\hat{C}$
0.05
0.04
0.03
0.02
0.01
0.00
0.00
0.01
0.02
0.03
0.04
0.05
injected label-correlated amplitude θ_{inj}